How to use Drager theatre ventilator on ITU patients

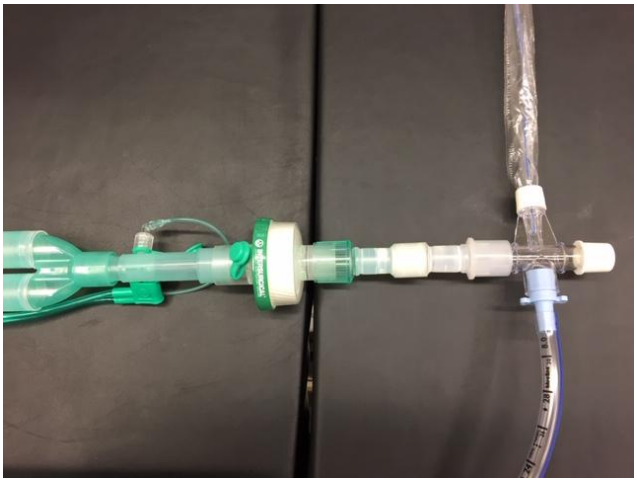
1 General

- An anaesthetist will be available in each surge ICU area
- Machines will be set up before use by ODPs
- Anaesthetic vaporizers and nitrous oxide cylinders and pipelines will have been removed
- A separate emergency ventilation bag must be available at all times. If you need to disconnect from anaesthetic machine to use it, remember to clamp tube.
- Always ensure that the manual breathing bag is full.
- Large leaks can be identified by an empty manual breathing bag. In this case take immediate action: Check for leakages, increase fresh gas flow.
- Man / Spont mode can be used for hand ventilation by adjusting the APL valve. Only do this if you have you been trained to do so. The APL valve has no influence when the machine is ventilating.
- If ventilator fails, machine will default to Man/Spont mode. Therefore always set APL valve either at spont or at desired PEEP level to avoid giving excess PEEP.



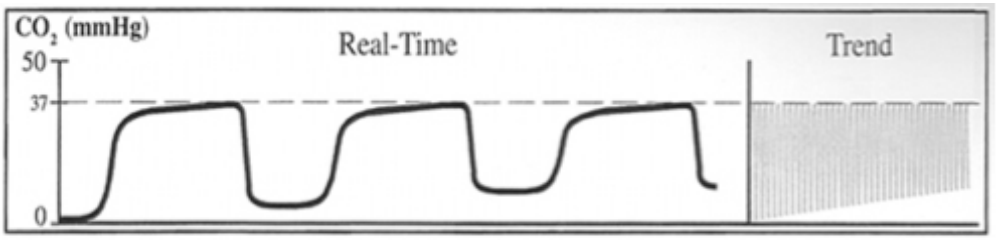
2 Set up

- Always use dry circuit
- Inspiratory arm: Fresh gas outlet – tubing – gas sampler – filter- in line suction- patient
- Expiratory arm: Patient – in line suction – filter – gas sampler - tubing – CO_2 absorber



3 CO_2 absorber (Soda Lime)

- Exhaled gases are reused in circle system. CO_2 is removed by soda lime (CO_2 absorber).
- An exhausted absorber can be identified by rising inspiratory CO_2. Make sure an inspiratory high CO_2 alarm is set and change soda lime when $FiCO_2$ reaches 0.5kPa. It can also be identified by a colour change of the soda lime (less than one third of the substance is still white).



4 Monitoring

- Sample gas line must always be connected. This is important to measure the FiO_2 and $EtCO_2$.
- Because of the circle system, the measured FiO_2 may differ from the set O_2 concentration on the fresh gas flow. (Make sure a low FiO_2 alarm is set)

5 Alarms

- Set alarm volume to 100%
- All alarm limits should be patient specific
- Make sure low FiO_2 alarm and high $FiCO_2$ alarms are set
- Set low airway pressure alarm (Paw) to be between PEEP and inspiratory pressure / plateau pressure. This allows detection of unintentionally applied CPAP as well as intrinsic PEEP situations.
- Alarm messages will disappear from the screen when the alarm condition has gone
- Check alarm history for missed past alarms

6 Flow

- Set the oxygen and air flow rates
- Total flow should be 1.5 X minute ventilation (RR x ideal Vt x 1.5). This will further minimise risk of re-breathing CO_2 and decrease excess humidity building up in circuit.

7 12 hour checks

- Every 12 hours check the following:
- The CO_2 absorber (as above)
- The water trap of the gas analyser. If more than half full, take it out and use a syringe (without needle) to draw the water out of the blue connector (it is post filter so not full of Covid – dispose down sink). Reinsert. If this doesn't work, ask ODP to replace.
- The hoses (tubing). Use hoses with water traps and empty these regularly.
- The filters. Change any that are wet or soaked.

8 72 hour checks

- Will be done by ODP or anaesthetist
- The patient must be disconnected prior to the test and ventilated by other means
- It may take up to 8 minutes
- Consist of a full device check including flow and volume measurement
- May need to be done before 72 hours if there are problems

9 Suctioning

- There is no volume compensation, so be as quick as possible.
- When not using the suction, turn it off
- After suctioning you may need to:
    - Increase fresh gas flow rate for a few minutes
    - Use the oxygen flush (carefully)

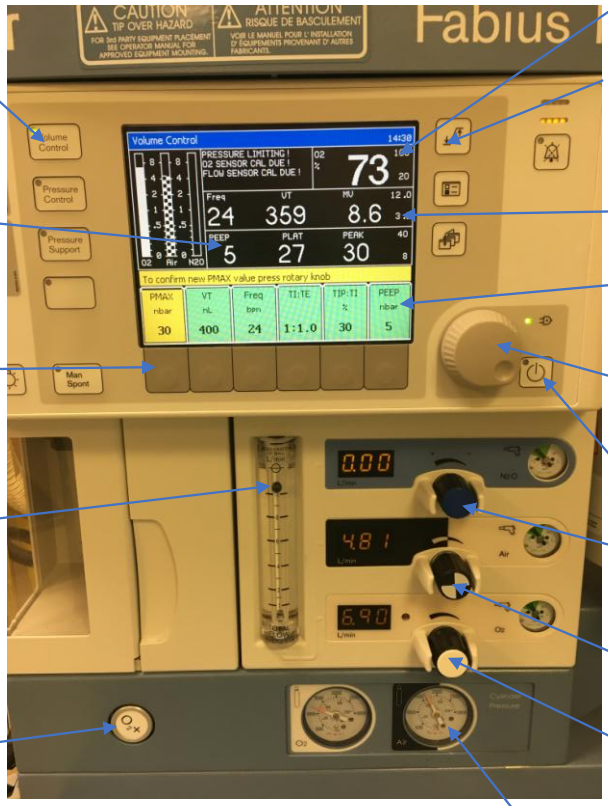
10 Nebulising

- Discuss with doctor as may interfere with ventilation
- Put nebuliser between filter and patient

11 Modes of ventilation available

- In volume control modes leakages will not be compensated for automatically. In this case, pressure controlled mode may be preferred.
- There is no apnoea ventilation in Pressure Support or Man / Spont modes

Type	Set up	Caution
Volume control	Vt; Resp Rate; PEEP Set Pmax to 30cmH_2O	No Pressure support. Spont breath not supported and might interfere with mandatory ventilation.
Pressure control	Pinsp; Resp Rate; PEEP Increase inspiratory flow rate to 50 litres/min	No pressure support. Spont breath not supported and might interfere with mandatory ventilation.
Pressure support	Pinsp; PEEP Can add pressure control ventilation at a set rate	No apnoea ventilation back up
Spont breathing without support (Man / Spont mode)	Bag ventilation possible	Check and adjust the APL valve No apnoea ventilation back up



Left hand buttons to change ventilatory mode

PEEP, Plateau pressure and Maximum inspiratory pressure

Press any button underneath to view and/or alter the above values

Total gas flow meter (aim for 1.5 x MV)

O2 flush. May be cautiously required after suctioning

Fraction of O_2 (%) in fresh gas flow (can be different to fraction of inspired O_2 if FGF is low)

Button to alter alarm settings

Respiratory rate, Tidal volumes and Minute ventilation

Rotary Control knob: to select, adjust, and confirm functions/settings

On/Off

N_20 flow control knob (not in use for COVID)

Air flow control knob

O_2 flow control knob

Pressure gauges for cylinder gases (should not be required in COVID set-up)

Quick guide when first connecting to a patient

- ODP / anaesthetist will have done machine check
- Check circuit looks like photo in “set up”
- Make sure separate emergency ventilation bag is available
- Turn APL valve to “Spont” or appropriate PEEP level
- Check alarm volume set to 100%
- Adjust alarm limits appropriately for patient. Of special importance are: FiO_2 low, $FiCO_2$ high, MV high / low, EtCO2 high / low.
- Set gas flow to 1.5 x minute ventilation. Start with 100% oxygen. Titrate down as required by decreasing the O_2 flow and increasing the air flow so the total remains at 1.5 x minute ventilation
- Set up chosen mode of ventilation. Remember to adjust Pmax when using volume control modes.